DesatNerd: Clinical & Technical User Guide

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1. Overview and Operational Purpose

DesatNerd is a signal processing application created to measure hypoxic burden metrics, define nocturnal desaturation morphology and extract physiological parameters from polysomnography (PSG).

Currently, sleep medicine relies heavily on the Apnoea-Hypopnoea Index (AHI) and Oxygen Desaturation Index (ODI). While helpful for gross categorical diagnosis, frequency metrics count each physiological collapse equally regardless of duration, nadir depth, or cumulative oxygen deficit. DesatNerd tried to overcome these limitations by performing the following:

- Reading multi-vendor polysomnography EDF recordings and standalone pulse oximetry streams (Compumedics, Nox, REMlogic, Nonin, Nellcor, Masimo).
- Synchronising PSG sleep staging annotations (`<study-name>.edf.XML`) to compute indices over Total Sleep Time (TST), Sleep Period Time (SPT), or Total Recording Time (TRT).
- Deploying multi-parameter physiological artefact screening to eliminate sensor disconnections, probe dropouts, and flatline freezes without distorting genuine hypoxic events.
- Quantifying classical hypoxaemia metrics (ODI 3%, ODI 4%, T90, T88, T85, T80).
- Calculating contemporary literature-validated hypoxic burden metrics:
    a. Azarbarzin Hypoxic Burden (%min/h)
    b. REDTA (Respiratory Event Desaturation Transient Area, %h and %min/h)
    c. Kulkas Desaturation and Recovery Severity (DesSev, RecSev, %min/h)

2. Supported File Formats and Parser Architecture

2.1 European Data Format (EDF / EDF+)

DesatNerd natively evaluates EDF/EDF+ signal headers and labels:

- **Signal Extraction:** Automatically identifies SpO2 channels across standard labels (`SpO2`, `SaO2`, `O2Sat`, `Oximeter`) and coupled Pulse Rate channels (`Pulse`, `PR`, `Heart Rate`).

- **Pure-Python Fallback:** Uses `pyedflib` when available and falls back to a 1992/2003 specification pure-Python binary reader to parse the 256-byte header and per-channel signal records.
- **Unit Scaling Correction:** Evaluates dynamic range. If SpO2 values are recorded as fractions (0.0 to 1.0), the engine automatically scales them to percentage points (0 to 100%).

2.2 Compumedics Profusion Companion XML Annotations

When analysing clinical recordings, DesatNerd searches the folder for matching companion XML files:

- **Candidate filenames:** `<study-name>.edf.XML`, `<study-name>.edf.xml`, `<study-name>_Profusion.xml`, and `<study-name>_events.xml`.
- **Sleep Staging:** Ingests `<SleepStages>` and maps Profusion stage codes (`0`/`10`=Wake, `1`=N1, `2`=N2, `3`/`4`=N3, `5`=REM) to 30-second epoch vectors.
- **Scored Events:** Extracts start times, durations, and concepts for obstructive apneas, central apneas, mixed apneas, hypopneas, and desaturations.

2.3 Nonin Pulse Oximeters

- Ingests ASCII text formats (`.asc`, `.csv`, `.txt`, `.dat`) produced by Nonin nVISION and WristOx2 devices.
- Handles variable header lengths, extracting date/time, SpO2, and heart rate.

2.4 Nellcor N-600x / OxiMax Series

- Nellcor serial ASCII outputs print recurring column headers (`TIME %SPO2 BPM PA Status`) every 25 lines and whenever alarm limits change.
- The parser cleans out banner metadata, strips repeated header text blocks, and filters out limit-violation asterisks (e.g., converting `190*` to numeric `190`).
- Translates missing-signal sentinels (`---`) directly into IEEE NaN representations.

2.5 Masimo Radical-7, Rad-97, and Rad-5

- Parses tabular exports and space-delimited TrendCom strings (`MM/DD/YY HH:MM:SS SpO2=097 PR=069 PI=04.69 EXC=800: SET`).
- Handles `000` and `---` exception flags, assigning true numeric channels for SpO2, Pulse Rate, and Perfusion Index (PI).

3. Signal Preprocessing & Multi-Parameter Artefact Detection

Accurate hypoxic burden calculation requires strict exclusion of non-physiological noise. Artefacts skew area integrals because artificial dropouts reach extreme nadirs (50% or lower) and linger over hundreds of samples.

3.1 Artefact Criteria in `detect_artifacts()`

The engine evaluates six distinct physiological constraints:

Constraint Type	Threshold / Limit	Operational Logic
Range Bounds	< 50.0% or > 100.5%	Readings outside valid range or non-numeric tokens are flagged immediately.
Slew-Rate Thresholds	> 4.0%/s (fall), > 6.0%/s (jump)	Gas exchange constraints limit true falls; faster rates indicate dropouts or adjustment.
Multi-Second Acceleration	> 10.0% over 3 seconds	Sustained steep drops exceeding physiological capabilities are flagged.
Flatline / Freeze Rejection	SD < 0.01 over 15s (SpO2 < 92%)	Detects low-perfusion or sensor lock states holding invariant values.
Pulse-Coupled Verification	< 25 bpm, > 240 bpm, > 40 bpm/s	Pulse anomalies or frozen pulse (SD < 0.01) invalidate concurrent SpO2 samples.
Island Bridging & Padding	< 3s duration, ±3.0s padding	Short valid segments between artefacts are bridged; borders receive padding.

3.2 Signal Smoothing in `preprocess_signal()`

- Identified artefact periods are replaced with NaN.
- Dropouts shorter than or equal to 5 seconds undergo linear interpolation.
- The interpolated array passes through a 1-second rolling median filter to suppress high-frequency hardware digitisation noise without flattening steep physiological desaturation slopes.

4. Oxygen Desaturation Detection & Morphology Engine

The `detect_desaturations()` function isolates discrete desaturation episodes and derives detailed kinematic and morphological metrics.

4.1 Baseline Determination

- **Rolling Quantile:** A 300-second-centred rolling 98th percentile window establishes the moving baseline.
- **Cyclical Ceiling Guard:** During severe repetitive sleep apnoea (e.g., 50 events/hr), standard rolling baselines can collapse under the weight of recurring desaturations. The engine pairs the 98th percentile baseline with a 90-second rolling maximum ceiling (`short_max - 0.5`), keeping the baseline stable during clustered events.

4.2 Candidate Envelope Identification

- A candidate drop begins when the signal falls 1.5% or more below the dynamic baseline.
- Artefact-masked samples are excluded from triggering new envelopes.

4.3 Start Point (Onset) Search

- From the trigger point, the engine looks back up to 45 seconds (constrained by the end of the previous event) to find the local maximum immediately preceding the fall. This peak defines the true pre-event baseline.

4.4 Nadir Isolation

- Scans forward from the true start to locate the lowest saturation sample.
- If an artefact occurs during the descent, the search terminates to avoid incorporating false dropouts.

4.5 Physiological Recovery Tracking

The engine avoids premature truncation by tracking resaturation using a two-tier approach:

- **Baseline Re-attainment:** Recovery ends when the signal returns to within 0.5% of the local baseline or dynamic baseline.
- **Crest / Inter-Event Boundary:** In incomplete recoveries (such as severe cyclical apnoea), the engine tracks the highest resaturation peak. If the signal rebounds by ≥ 50% of the drop depth (or ≥ 2.0% absolute) and subsequently rolls over by ≥ 1.0% from that recovery peak, the recovery terminates at the crest. That point marks the end of the current event and the potential baseline for the next.

4.6 Morphological Parameters

For every validated event, DesatNerd records:

- **Duration (s):** Total time from onset to recovery.
- **Fall Duration & Recovery Duration (s):** Split at the nadir timestamp.
- **Depth (%):** Difference between local baseline and nadir.
- **Fall Slope (%/s):** Depth / Fall_Duration
- **Recovery Slope (%/s):** Recovery SpO2 - Nadir / Recovery Duration

- **Segment Areas (%·sec and %·min):** Calculated separately for the falling and recovery limbs using numerical integration.

5. Mathematical Formulations & Hypoxic Burden Metrics

5.1 Conventional Metrics

- **ODI (Oxygen Desaturation Index, events/h):** Number of desaturations meeting the threshold (3% or 4%) divided by analysis duration in hours.
- **T90, T88, T85, T80 (% and min):** Cumulative duration spent below 90%, 88%, 85%, and 80% SpO2, expressed in total minutes and as a percentage of valid, artefact-free analysis time.

5.2 Azarbarzin Hypoxic Burden (%min/h)

- **Reference:** Azarbarzin et al., *European Heart Journal* (2019).
- **Linked Respiratory Events:** Integrates the area between the pre-event baseline (100-second maximum before event end) and the SpO2 curve in a search window from -5 s to +45 s relative to event termination. Overlapping windows are capped to prevent double-counting.
- **Standalone Mode:** When scoring without respiratory annotations, the engine sums the total trapezoidal desaturation area across all identified events.
- **Normalisation:** Total area in %·min divided by analysis hours.

5.3 REDTA (Respiratory Event Desaturation Transient Area)

- **Reference:** de Chazal et al., *Sleep* (2021).
- **Formulation:** Evaluates an analysis window starting at the respiratory event midpoint and extending for 2.5 times the event duration.
- **Baseline Reference:** Integrates desaturation transients against a fixed 100.0% baseline: Area = $\int (100.0 - \mathrm{SpO2}(t))\, dt$
- **Output:** Expressed as total percent-hours (%h) and normalised hourly rate (%min/h).

5.4 Kulkas Severity Framework

- **Reference:** Kulkas et al., *Medical & Biological Engineering & Computing* (2013).
- **Desaturation Severity (DesSev, %min/h):** Sum of the falling phase areas divided by analysis hours. Reflects speed and depth of collapse.
- **Recovery Severity (RecSev, %min/h):** Sum of the recovery phase areas divided by analysis hours. Reflects ventilatory drive and post-event restitution.
- **Total Severity (%min/h):** DesSev + RecSev

5.5 Analysis Windows: TST vs. SPT vs. TRT

- **Total Sleep Time (TST):** Constrains analysis strictly to 30-second epochs scored as sleep (N1, N2, N3, REM).
- **Sleep Period Time (SPT):** Evaluates from sleep onset (first sleep epoch) to sleep offset (last sleep epoch). Includes intervening wakefulness (WASO). Recommended because desaturations terminating in brief arousals remain accounted for.
- **Total Recording Time (TRT):** Evaluates the entire recording from lights-off to lights-on. Used for standalone oximetry without PSG staging.

5.6 Numerical Integration Quadrature

- **Trapezoidal Rule:** First-order closed Newton-Cotes formula connecting discrete data points with linear segments.
- **Simpson's Rule:** Second-order Newton-Cotes formula fitting parabolic arcs across sample triads to capture curvature near desaturation nadirs.

6. Graphical Interface & Clinical Workflow

6.1 Interactive Oximetry Trace

- **Preset Windows:** Inspect data across Full Recording, 1 Hour, 15 Minutes, 5 Minutes, 1 Minute, or 30 Seconds (1 Epoch).
- **Dual-Plot Canvas:**
    - **Upper Detailed Axis:** Displays the SpO2 trace, moving baseline, shaded orange desaturation areas, red nadir dots, and a top hypnogram banner coloured by sleep stage (Wake=Grey, N1=Light Blue, N2=Mid Blue, N3=Dark Blue, REM=Green).
    - **Artefact Visualisation:** Artefact periods are shaded with light red hatching (`#e74c3c`), providing immediate visual confirmation of excluded data.
    - **Lower Overview Axis:** Displays the full night minimap with a moveable orange viewport rectangle.
- **Event Linking:** Clicking any event row in the morphology table centres the viewer directly on that desaturation in a 5-minute window.

6.2 Key Metric Cards

The dashboard displays 12 summary cards covering duration, mean, minimum, baseline SpO2, ODI (3% and 4%), T90 (% and min), Azarbarzin HB, REDTA rate, Kulkas DesSev, and kinematic slopes.

6.3 Batch Processing

Select a directory containing recordings to process files in batch mode. The queue table displays real-time completion status and automatically appends results to an aggregated CSV spreadsheet in the output folder.

7. Step-by-Step Operating Instructions & Clinical Workflow

7.1 Launching the Application & Workspace Layout

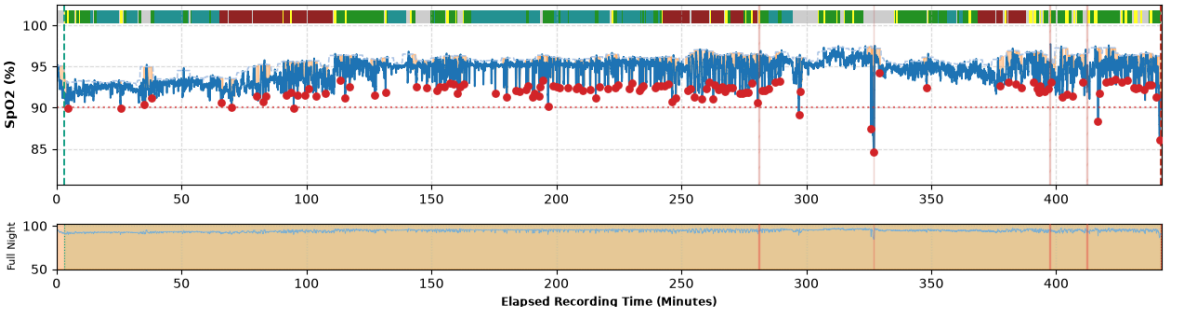
- **Starting DesatNerd:** Run `python main.py` or double-click the compiled binary (`DesatNerd.exe`).
- **Workspace Organisation:**
    - **Left Sidebar:** Parameter setup, input selection, batch mode toggles, execution triggers, and the `User Guide` button.
    - **Top Cards:** Real-time display of the 12 primary clinical metrics with threshold alert coloring.
    - **Main Tabbed View:** Interactive Trace Viewer, Desaturations & Morphology Table, Literature Comparison Table, and Batch Queue.
    - **Bottom Console:** System log recording file detection, XML parsing, and processing status.

7.2 Single Study Analysis Workflow

1. **Select the Input File:** Click Browse File in the left sidebar and choose your recording (`.edf`, `.csv`, `.asc`, etc.). DesatNerd automatically checks for companion XML annotation files in the same directory.
2. **Set the Output Directory:** Target the folder where summary tables and event logs will be saved.
3. **Configure Parameters:**
    - Select the Desaturation Threshold (3% or 4%).
    - Choose the Baseline Window (300s Standard, 180s, or 600s).
    - Choose the Analysis Time Window (SPT, TST, or TRT).
    - Select the Integration Quadrature (Trapezoidal Linear vs. Newton-Cotes Simpson).
4. **Execute Analysis:** Click START ANALYSIS. Monitor progress in the progress bar and log console.

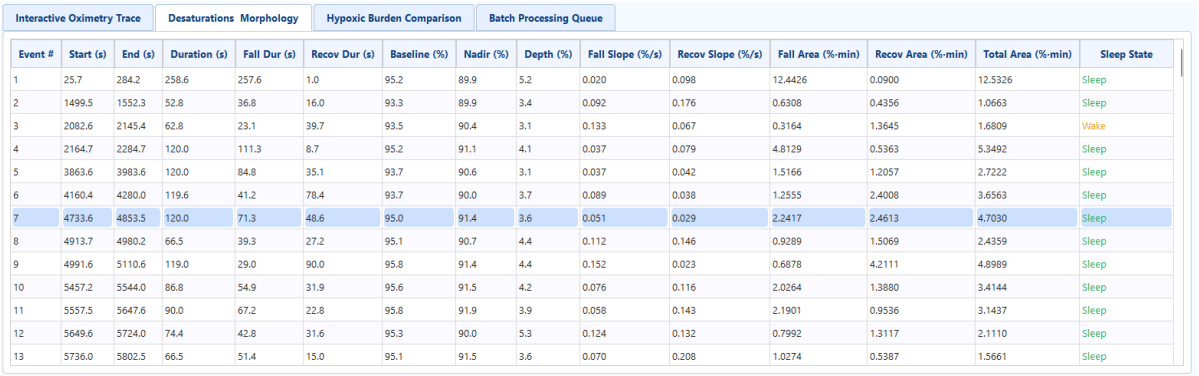
7.3 Reviewing the Interactive Trace Viewer

- **Inspecting Signals:** View the blue SpO2 line, dashed baseline, orange desaturation fills, red nadirs, and light red hatched artefact bars.
- **Changing Scale:** Select intervals from 30 seconds up to full recording.
- **Panning:** Use Prev Window / Next Window, the continuous pan slider, or click directly on the bottom minimap to jump anywhere in the study.
- **Window Metadata:** Check the status banner above the plot for time span, epoch numbers, active sleep stage, and SPT status.

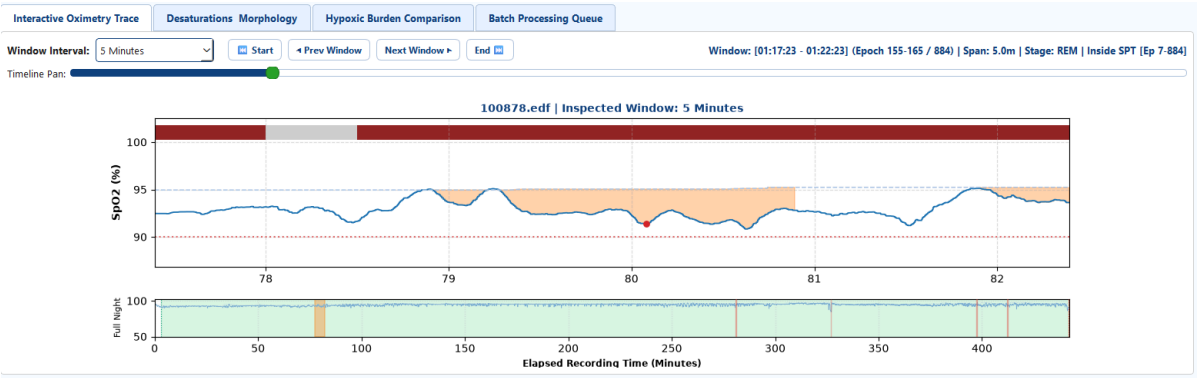


7.4 Reviewing Morphology and Literature Benchmarks

- **Event Table:** Click any row in the Desaturations & Morphology tab to centre the trace viewer on that specific event in a 5-minute window.
- **Literature Comparison:** View calculated Azarbarzin HB, REDTA, and Kulkas scores alongside published CVD mortality cut-offs.



Event #	Start (s)	End (s)	Duration (s)	Fall Dur (s)	Recov Dur (s)	Baseline (%)	Nadir (%)	Depth (%)	Fall Slope (%/s)	Recov Slope (%/s)	Fall Area (%-min)	Recov Area (%-min)	Total Area (%-min)	Sleep State
1	25.7	284.2	258.6	257.6	1.0	95.2	89.9	5.2	0.020	0.098	12.4426	0.0900	12.5326	Sleep
2	1499.5	1552.3	52.8	36.8	16.0	93.3	89.9	3.4	0.092	0.176	0.6308	0.4356	1.0663	Sleep
3	2082.6	2145.4	62.8	23.1	39.7	93.5	90.4	3.1	0.133	0.067	0.3164	1.3645	1.6809	Wake
4	2164.7	2284.7	120.0	111.3	8.7	95.2	91.1	4.1	0.037	0.079	4.8129	0.5363	5.3492	Sleep
5	3863.6	3983.6	120.0	84.8	35.1	93.7	90.6	3.1	0.037	0.042	1.5166	1.2057	2.7222	Sleep
6	4160.4	4280.0	119.6	41.2	78.4	93.7	90.0	3.7	0.089	0.038	1.2555	2.4008	3.6563	Sleep
7	4733.6	4853.5	120.0	71.3	48.6	95.0	91.4	3.6	0.051	0.029	2.2417	2.4613	4.7030	Sleep
8	4913.7	4980.2	66.5	39.3	27.2	95.1	90.7	4.4	0.112	0.146	0.9289	1.5069	2.4359	Sleep
9	4991.6	5110.6	119.0	29.0	90.0	95.8	91.4	4.4	0.152	0.023	0.6878	4.2111	4.8989	Sleep
10	5457.2	5544.0	86.8	54.9	31.9	95.6	91.5	4.2	0.076	0.116	2.0264	1.3880	3.4144	Sleep
11	5557.5	5647.6	90.0	67.2	22.8	95.8	91.9	3.9	0.058	0.143	2.1901	0.9536	3.1437	Sleep
12	5649.6	5724.0	74.4	42.8	31.6	95.3	90.0	5.3	0.124	0.132	0.7992	1.3117	2.1110	Sleep
13	5736.0	5802.5	66.5	51.4	15.0	95.1	91.5	3.6	0.070	0.208	1.0274	0.5387	1.5661	Sleep



7.5 Batch Processing Workflow

1. Check Enable Batch Processing Mode in the sidebar.
2. Browse and select the folder containing your cohort files.
3. Inspect the populated queue and click START ANALYSIS.
4. Results are saved to desatnerd_batch_summary.csv and individual event files.

7.6 Troubleshooting Common Issues

- **Missing Sleep Stages:** Verify the companion XML file sits in the same directory and shares the exact base name with the EDF.
- **Disconnected Pulse Signals:** Ensure pulse rate tracks are genuine; non-physiological pulse leads are safely ignored by the guard.
- **Log Inspection:** Review Python error tracebacks in the bottom diagnostics console if a study fails to process.

8. Recommended Diagnostic Thresholds

Metric	Category	Range / Threshold	Clinical Significance
ODI (3% or 4%)	Normal	< 5.0 events/h	Within normal physiological limits.
	Mild	5.0 – 14.9 events/h	Mild nocturnal respiratory disturbance.
	Moderate	15.0 – 29.9 events/h	Moderate oxygen desaturation frequency.
	Severe	≥ 30.0 events/h	Substantial desaturation frequency.
T90	Hypoxaemic	> 10.0% of sleep time	Clinically significant nocturnal hypoxaemia.
Azarbarzin Hypoxic Burden	High Risk	> 53.0 – 60.0 %min/h	Correlates with heightened cardiovascular mortality independently of AHI.